A Lamb Without Blemish:

Demand for Sheep Coat Color in Sierra Leone

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Abstract:

In this paper, we analyze demand for sheep coat color among sheep-owning households in Sierra Leone. In West Africa, sheep play an important role in religious ceremonies which some studies claim makes white-coated sheep more valuable. We test this hypothesis by designing a discrete choice experiment which uses scientific illustrations to vary the weight and coat color of a representative West African Dwarf sheep, the most common breed in Sierra Leone. We find that respondents were willing to pay about twice as much for a sheep with a pure white coat compared to the more common black and white sheep. Sheep with only a few black spots are valued the same as sheep with far less white in their coats, underscoring the importance of sheep being pure white. Our results suggest that cultural preferences for coat color could be an important factor in whether households adopt new sheep breeds.

Introduction

Livestock are an important component of food security strategies. In addition to being a source of calories for the household, they are seen as high value assets that can be a store of value and a high-income farming activity. Thus, a great amount of effort goes into breeding programs to improve the productive ability of livestock. For example, breeding programs in some places are focusing on improving weight gain, milk production, and disease resistance to make livestock-rearing a more productive activity for the household (Ojango et al. 2021; Rege et al. 2011; Gutu et al. 2015).

However, another function of livestock is often ignored in these breeding programs: their role in cultural and religious functions. In West Africa, Muslim households typically slaughter a sheep for the holiday Eid al-Adha, a celebration of Abraham's sacrifice. Sheep are also typically slaughtered for weddings, funerals and baby naming ceremonies. Because of the role they play in cultural and religious holidays, some have suggested that pure white sheep are more valuable than black and white sheep. One study claims that, because of these cultural and religious considerations, white sheep are three-times more valuable than black sheep in West Africa for religious reasons (Traoré et al. 2012). If this is true, this has important implications for breeding programs, as this may mean that breeding efforts

should take these preferences into account when producing improved breeds for households.

We test the hypothesis that sheep with pure white coats are significantly more valuable than sheep with black in their coats by conducting a discrete choice experiment for sheep owning households. In Sierra Leone, the most common breed of sheep is the West African Dwarf sheep which has a coat that is usually some combination of black and white (Bemji et al. 2012). Using scientific illustrations, we experimentally vary coat color and weight to determine whether there is a premium for sheep with a pure white coat. Since a large portion of our population is illiterate, our commissioned scientific illustrations are a crucial part of helping these households understand the choices. By making all the information visually apparent to the participant, we maximize the odds that participants are informed about what they are choosing.

We find that that pure white sheep are significantly more valuable than the more common black headed sheep found among the breed. Interestingly, we find that even sheep that are all white but have a few black spots are viewed as just as valuable as sheep that have more black in their coat. Thus, respondents are unlikely to be valuing coat color for its heat and disease resistance since the sheep with a few black spots are functionally as resistant as the pure white sheep. Our results are broadly consistent with this conclusion when using a mixed logit model and a latent class model.

Our paper makes a significant contribution to the literature on the valuation of livestock characteristics in the context of poor rural households. Several studies study the influence of coat color on the valuation of animals, with most studies finding that white coats are preferred and black coats are not (Kassie et al. 2023; Terfa et al. 2013; Gizaw et al. 2011; Zanou et al. 2024). The aversion to black coats, however, has been explained not as a religious or cultural preference but as an adaptation to heat stress and disease resistance. In particular, the tsetse fly has been found to be more attracted to black-coated animals, making black-coated animals more prone to *trypanosomiasis*, or sleeping sickness (Steverding and Troscianko 2004; Saini et al. 2017).

Our study helps disentangle the cultural and religious preference for whiteness from the aversion to black-coated animals for productive purposes by experimentally varying coat color from white to black. While adding black spots to an otherwise white sheep would not significantly change its heat or disease tolerance, it may change the cultural value by making it not pure white. By varying the coat color this way, we find evidence that a pure white coat carries a premium in the Sierra Leone sheep market. This is consistent with a cultural preference for whiteness that is not explained by any productive traits.

Our results carry some important implications for policy. Breeding programs have often focused on enhancing productive traits in animals with the view that these are the primary determinants of their value. If there are significant cultural preferences for sheep traits unrelated to production, new breeds produced by production-focused programs may not have high uptake among households. Our work here finds evidence for such cultural

preferences in Sierra Leone and suggests that future work should be studying these preferences more closely to potentially improve uptake of new breeds in the future. These results also highlight the importance of incorporating cultural considerations into the development of new agricultural technology.

Literature Review

There is a large literature studying demand for animal traits in low-income countries. Demand for animal traits is a salient topic of study because many breeding programs are trying to produce improved breeds for use by households. Breeding programs are necessary because foreign, often European, breeds will not perform well in tropical environments and may be vulnerable to diseases. Thus, breeding programs aim to make new breeds that combine the productivity of European breeds with the resilience of indigenous breeds.

Valuing the traits of indigenous breeds, sometimes called animal genetic resources (AGR), is determined through either hedonic analysis of livestock transactions (Jabbar and Diedhiou 2003; Scarpa et al. 2003; Ayele et al. 2006; Zanou et al. 2024) or choice experiments conducted among households, traders, or breeders (Ruto et al. 2008; Ouma et al. 2007; Kassie et al. 2009; 2023). In these analyses, it is generally found that traits like weight gain, milk production, and disease resistance are important determinants of these valuations.

Yet, choice experiments and hedonic analyses often find that coat color and appearance matter for sheep and goats. In particular, it is a common finding in African contexts that sheep with black coats have lower prices (Terfa et al. 2013; Ayele et al. 2006; Duguma et al. 2011; Kassie et al. 2023; Tadesse 2009; Gizaw et al. 2011). However, the avoidance of black coats has been explained as not an aesthetic preference but instead an adaptation to heat stress and disease. Black-coated animals are more prone to heat stress since their coats absorb more sun and it has been found that tsetse flies which can bite animals and cause *trypanosomiasis* are more attracted to dark surfaces (Steverding and Troscianko 2004; Saini et al. 2017).

Similarly, studies have that sheep and goats with brown or white coats are priced higher holding other traits constant (Kassie et al. 2023; Tadesse 2009; Kassa et al. 2011; Zanou et al. 2024). Some of these studies speculate that coat color is important for religious and cultural reasons and not solely as a proxy for disease or heat tolerance (Ouma et al. 2007; Terfa et al. 2013; Ibrahim et al. 2019). In the case of West Africa, Traoré et al. (2012) claims that "due to religious and cultural reasons, consumers pay a major attention on the coat colour of the sheep as it used for Tabaski," where Tabaski is another name for the Muslim holiday Eid al-Adha. Sheep play a specific role in Eid al-Adha, the "festival of sacrifice" which celebrates Abraham's sacrifice of a sheep instead of his son Ishmael in Muslim tradition. Traditionally, sheep are used in West Africa though cattle and goats are also common sacrifices.

Traoré et al. (2012) also claims that the white sheep have a three-times higher selling price than black sheep. In the case of goats in southern Ethiopia, Kassie et al. (2023) finds a 50-60% price increase for white coat color in goats using a choice experiment design. In a hedonic analysis of sheep markets in Ethiopia, Tadesse (2009) finds only a 13% premium for white coats in central Ethiopia and Kassa et al. (2011) finds an 8-16% premium in northeastern Ethiopia. Similarly, Zanou et al. (2024) finds only a 3% price premium for white coat color in goats in Benin but no premium for sheep.

It is difficult to understand whether preference for white sheep in a country like Sierra Leone is due to cultural or religious reasons or whether there are reasons that valuing whiteness is related to the productive ability of the sheep. The West African Dwarf sheep, the most popular sheep breed in Sierra Leone, present an opportunity to study the influence of whiteness of sheep on how households will value them. West African Dwarf sheep can be all white, all black, and some mixture of white and black. One common iteration is for the sheep to have a black head with a white body (Bemji et al. 2012).

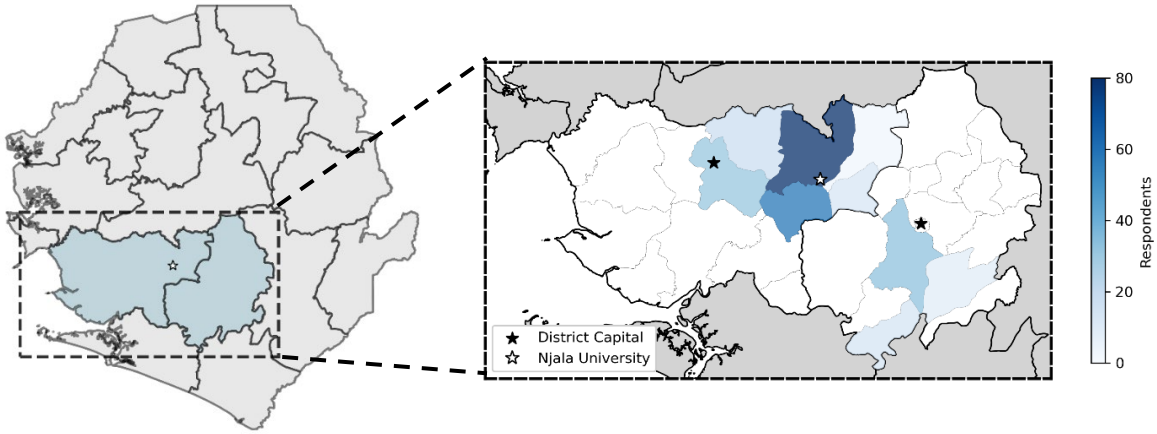
In our paper, we design a choice experiment to explicitly test for the influence of whiteness on the value of sheep in Sierra Leone, a majority Muslim country. A key part of this study is commissioning scientific illustrations of five different color combinations, all varying degrees of black and white, which will be used to see the influence of small amounts of black on an otherwise white sheep. Of particular interest is the difference between a white sheep and a white sheep *with black spots*. While the black spots will not significantly impact its heat and disease tolerance, it will make the sheet not "pure white" anymore. If we see black spots causing a significant change in the valuation, this is evidence that whiteness matters for a reason other than heat or disease tolerance.

Data

To test our hypothesis, we collected a convenience sample of 275 sheep-owning households in the Moyamba and Bo districts of Sierra Leone. Our enumerators were graduate students in animal science based out of Njala University and were asked to survey sheep-owning households in either of the districts. The geographic distribution of the sample is plotted in Figure 1.

Table 1 reports the summary statistics of our 275 households, both characteristics of the respondent and of the household. About 50% of respondents are between 40 and 60 years of age, 35% are female, and 33% have no formal education. Moyamba and Bo are in the south of Sierra Leone and so tend to be majority Mende and Temne (about 80% of our respondents), two common ethnic groups. The Fula ethnic group (about 9% of our sample), usually located in the north, are specifically known for raising livestock while Mende and Temne are not known for having experience with livestock (Adebayo 1991). About 70% of our respondents are Muslim while about 30% are Christian, which is slightly more Christian representation than the national average in 2023 (77% Muslim and 22% Christian) (US State Department 2023).

Figure 1: Study Area in Sierra Leone



Most households in our sample make between 500k and 1,000k leones, or between 25-45 USD, in a single month. Considering that the average household size is 5.5 individuals, this implies that most individuals in our sample are living on much less than 1 USD per day. Our survey asks the respondents to report all of their assets, which we then aggregate into an index using Principle Component Analysis (PCA). Similarly, our survey asks questions about animal husbandry practices which we similarly aggregate into one index using PCA. Positive values of the asset index indicate a larger number of assets owned whereas positive values of the husbandry index indicate good husbandry practices such as removing manure from pens, practicing rotational gazing, and having access to an animal health worker.

About 83% of our sample reported income from livestock, and on average a household kept 6.6 sheep. Households were also asked whether they specifically kept sheep for the Eid al-Adha holiday, and 47% indicated that they did. However, a larger portion of respondents indicated that they kept sheep for income from sales (80%), manure (71%), and other ceremonies and other holidays (66%). About 59% indicated that they kept sheep as a form of savings or insurance. Thus, keeping sheep for the purposes of slaughtering on a holiday or ceremony is quite common in our sample, though keeping it for income is the most common.

Our survey also asks respondents to indicate their preference for sheep coat color and the traits of a breeding ram. The results from these questions are displayed in Table 2. When asked which color of sheep they prefer between white, black and white, and black, about 75% of respondents chose white. Only about 1% of respondents chose black. When asked which ram traits they would consider when breeding, size was the most widely considered (87%). Yet, 64% of respondents considered ram color and 27% of respondents considered it the most important ram trait (compared to the most common answer, testicles, at 33%).

Table 1: Summary Statistics

	Obs	Mean	Std. Dev.
Respondent			
Education	275		
No Schooling		0.33	0.47
Some Primary Schooling		0.17	0.38
Completed Primary		0.28	0.45
Completed Secondary		0.21	0.41
Age (Years)	275		
< 30		0.11	0.31
30-40		0.19	0.40
40-50		0.26	0.44
50-60		0.25	0.43
> 60		0.19	0.39
Female (Yes/No)	274	0.35	0.48
Ethnicity	275		
Mende		0.71	0.45
Temne		0.11	0.31
Fula		0.09	0.28
Other		0.09	0.29
Religion	269		
Muslim		0.69	0.46
Christian		0.30	0.46
Other		0.01	0.09
Household			
Monthly Income	260		
< 500k		0.19	0.39
500k-1,000k		0.38	0.49
1,000k-1,500k		0.22	0.41
> 1,500k		0.21	0.41
Household Size	275	5.55	4.49
Asset Index	268	0.00	2.66
Husbandry Index	275	0.00	1.40
Livestock Income (Yes/No)	276	0.83	0.38
Size of Sheep Herd	275	6.16	5.36
Keeps Sheep for Eid (Yes/No)	275	0.47	0.50

Table 2: Preference for Sheep Traits

Color Traits				
	Frequency (%)			
White	74.62			
Black & White	23.48			
Black	1.52			
Ram Traits				
		Ranking		
	Select All	*First*	*Second*	*Third*
	%	%	%	%
Testicles	45.82	33.07	1.98	0.83
Size	86.91	28.74	46.03	13.28
Color	64.36	27.17	13.49	18.26
Age	69.45	7.48	15.48	22.41
Activeness	74.91	2.76	21.03	41.91
Defects	12.36	0.00	0.79	2.90
Feet	16.73	0.79	1.19	0.41

Methodology

Our method for understanding the willingness-to-pay for coat color in sheep is a dichotomous choice experiment. To keep the choices simple, we chose each option to have only three attributes: weight, color, and price. In our field observations, we judged these to be the most salient attributes for households when purchasing sheep. Since males are the most common to buy for slaughter, the sheep choices are all male. The levels of the choice attributes are shown in Table 3.

Since a significant portion of our population is illiterate, we aimed to have both coat color and weight to be discerned from one illustration. To this end, we commissioned scientific illustrations of sheep which had three different weights and five different coat colors. The three levels of weight are based off Body Condition Scores, a score used by veterinarians to appraise the condition of an animal, typically by feeling different parts of the body (Thompson and Meyer 1994). When designing the levels, the scientific illustrator used body conditions 2 ("thin"), 3 ("average"), and 4 ("fat"). When administering the experiment, these weight levels are referred to as "below average weight", "average weight", and "above average weight."

To choose coat color options, we combined observations in the field of the most common coat types along with the coat types discussed in Bemji et al. (2012). We also chose these coat colors to represent a gradual gradient from black to white. Our reference coat pattern is a sheep with a white body and black head, which field observations determined was a very common coat pattern. On the furthest ends, we have pure white and pure black sheep.

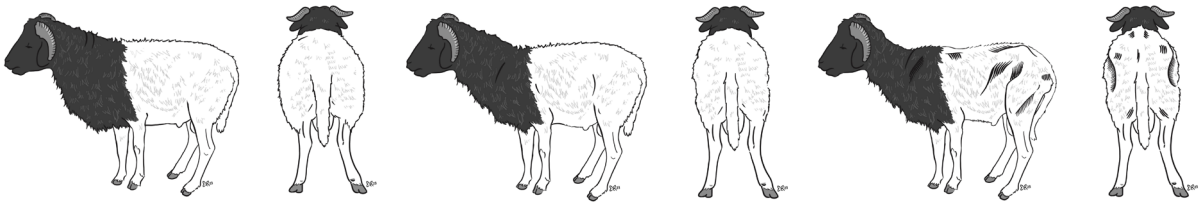
We then chose one sheep to be pure white except for some black spots, which is a common coat pattern and ideal for testing the significance of a pure white sheep. The coat color pattern one step removed from black is a sheep with a black head and partly black body, which is a common iteration that has marginally more black than the black head with white body.

Lastly, we collected a small sample of prices from a sheep market in the city of Bo, which is the capital of the Bo district, to determine our price levels. The cheapest price is 1,000 leones, or about 43 USD, and the price levels increased in 500 leon increments until reaching 3,000 leones (about 130 USD). In another part of the survey, we asked about sheep purchases and find evidence that a household typically never paid more than 2,000 leones for a single sheep or goat.

Table 3: Choice Experiment Attribute Levels

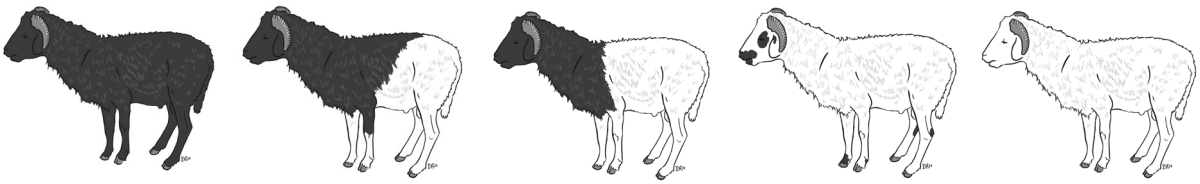
Weight

"Above Average Weight" Body Score: 4	"Average Weight" Body Score: 3	"Below Average Weight" Body Score: 2



Color

"Black"	"Black and White"	"White with Black Head"	"White with Black Spots"	"White"



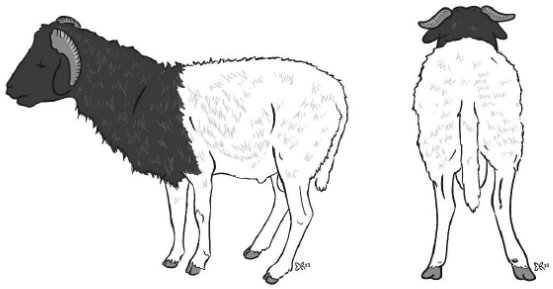
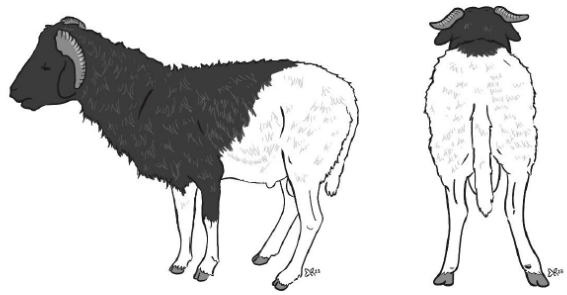
Price Levels (thousand leones)

1,000 1,500 2,000 2,500 3,000

In the choice experiment, respondents were presented with dichotomous choices with an opt-out option ("Neither"). In the administering of the survey, the enumerators read out the characteristics of each choice and showed the respondent the images. An example of a choice card is seen in Figure 2. When designing the choices, we chose 20 total pairs to

maximize our D-efficiency score. We then broke the 20 total pairs into four blocks, leaving five choices for each respondent.

Figure 2: Choice Card Example

Sheep 1	Sheep 2
C061 	**C062** 
This sheep costs **1,000 Leones**	This sheep costs **2,000 Leones**
This sheep is an **average weight**	This sheep is an **average weight**
This sheep is **white with a black head**	This sheep is **black and white**

Q1203C: The respondent chooses:

- ❑ Sheep 1
- ❑ Sheep 2
- ❑ Neither

To determine the willingness-to-pay for each characteristic, we first estimate a conditional logit model and then a mixed logit model. Using the random utility framework (McFadden 1973), the utility respondent i receives from choosing alternative j in choice task t can be defined as $U_{ijt} = V_{ijt} + \varepsilon_{ijt}$. Here, V_{ijt} is the systematic portion of the consumer's utility and ε_{ijt} is the random and unobserved portion. If we assume the error term is independent and identically distributed according to the type I extreme value, we can estimate the probability of consumer i choosing alternative j in choice task t by integrating over the values of ε_{ijt}:

$$\Pr\left(U_{ijt} > U_{ikt} \ \forall\, k\right) = \frac{e^{V_{ijt}}}{\sum_{k=1}^{J} e^{V_{ikt}}}$$

We parameterize the utility V_{ijt} as a function of the attributes of each choice:

$$V_{ijt} = \alpha + \beta Price_{ijt} + \sum_{w=1}^{W} \gamma_w weight_{ijt}^{w} + \sum_{c=1}^{C} \delta_c color_{ijt}^{c}$$

Where $weight_{ijt}^{w}$ and $color_{ijt}^{c}$ are fixed effects for each of the levels of the weight and color attributes. As a conditional logit, we estimate the likelihood conditional to each respondent, effectively controlling for the characteristics of the individual.

A weakness of conditional logit is that it assumes the irrelevance of independent alternatives (IIA). To test the robustness of our result, we also estimate the model as a random parameter (mixed) logit model which models each coefficient on each level of the characteristic as a random variable. The price parameter β we model as lognormal distributed and the other coefficients γ and δ as normally distributed. We use 1,000 Modified Latin Hypercube Sampling (MLHS) draws in the process of estimation.

As a robustness check, we also explore the potential influence of respondent heterogeneity on our results. Of special interest is whether the respondent's religion, particularly being Muslim, has an impact on the valuation of pure white sheep. This could be due to the use of sheep in Muslim holidays and ceremonies, especially the holiday Eid al-Adha (Traoré et al. 2012). We investigate this hypothesis in two ways. First, we interact the Muslim indicator with color in the conditional logit to see whether Muslim respondents value a pure white coat differently than non-Muslims. To consider many more dimensions of heterogeneity, we also estimate a latent class model (LCM) which allows the data to define classes based on respondent characteristics, including religion. We present the results of the LCM in the Appendix.

Results

In terms of interpreting the results, we note that only three respondents chose to opt out of the choice and this respondent was dropped from analysis. This impacts the interpretation of the results as being conditional on wanting to purchase a sheep, and so only representative of households that already want to purchase a sheep.

The results of a logit model and our two conditional logit models, one without a Muslim interaction and one with an interaction, are presented in Table 4. The results in all three models are quite similar. Price has the expected, negative coefficient and respondents value the sheep less if it has lower average weight and higher if it has an above average weight. Yet, the penalty for having a sheep be below average weight is much larger than the sheep being above average weight (about three times more).

In terms of coat color, our results agree with previous literature and show that respondents do not prefer black sheep. Rather, white sheep are valued at about double the "base" sheep which has a white body and a black head: the white sheep premium is about 0.8 compared to the constant of 0.86 in the logit model. A sheep that is white with black spots is valued the same as the base sheep, as the coefficient for this category is not statistically significant. This means that simply adding black spots to a white sheep significantly impacts its valuation by the respondents. In other words, this is evidence that a pure white coat carries a significant premium. In terms of willingness-to-pay, the pure white color premium is about 2,811.85 leones (0.807/-0.000287). However, this estimate of willingness-to-pay is sensitive to the estimation of the price coefficient.

In column 3 of Table 4, we add an interaction between the coat color and a binary variable indicating that the respondent is Muslim. None of the interactions with Muslim are

Table 4: Logit and Conditional Logit

	Logit	Conditional Logit	
	(1)	(2)	(3)
Price	-0.000302***	-0.000287***	-0.000291***
	(0.0000609)	(0.0000586)	(0.0000588)
Color (Base: White, Black Head)			
White	0.868***	0.807***	0.809***
	(0.133)	(0.127)	(0.226)
White w/ Black Spots	0.267*	0.182	0.129
	(0.136)	(0.136)	(0.235)
Black and White	0.340*	0.344*	0.552*
	(0.140)	(0.143)	(0.251)
Black	-1.260***	-1.144***	-1.178***
	(0.150)	(0.149)	(0.262)
Weight (Base: Avg. Weight)			
Below Avg. Weight	-1.427***	-1.262***	-1.259***
	(0.107)	(0.102)	(0.102)
Above Avg. Weight	0.455***	0.417***	0.421***
	(0.120)	(0.117)	(0.117)
Color x Muslim			
White			-0.286
			(0.999)
White with Black Spots			-0.205
			(0.999)
White with Black Head			-0.282
			(1.006)
Black and White			-0.595
			(1.000)
Black			-0.232
			(1.005)
Constant	0.864***		
	(0.176)		
Observations	2730	2730	2730
Pseudo R^2	0.144	0.165	0.166

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 5: Mixed Logit Results

		Estimate
Constant	Mean	0.588***
		(0.208)
	Std. Dev.	1.508***
		(0.342)
Price	Mean	-4.172**
		(1.887)
	Std. Dev.	3.425***
		(1.111)
Weight (Base: Avg. Weight)		
Below Avg. Weight	Mean	-1.339***
		(0.319)
	Std. Dev.	1.451***
		(0.330)
Above Avg. Weight	Mean	0.541*
		(0.311)
	Std. Dev.	0.390
		(0.363)
Color (Base: White, Black Head)		
White	Mean	1.160***
		(0.284)
	Std. Dev.	1.283***
		(0.412)
White with Black Spots	Mean	0.311
		(0.327)
	Std. Dev.	0.934
		(0.748)
Black and White	Mean	0.200
		(0.260)
	Std. Dev.	0.875
		(0.923)
Black	Mean	-2.816***
		(0.687)
	Std. Dev.	3.079***
		(0.684)
Observations	2,730	

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

significant, indicating that the valuation of coat color does not depend on the respondent's religion. While sheep are used extensively in Muslim holidays, the preference for pure white sheep may extend beyond these specific uses. Moreover, if the sheep's market value is higher due the coat color's importance to ceremonies, Christian respondents may still value coat color because they know the sheep will have resale value if used for breeding.

Table 5 presents the results of the mixed logit model. For the model to converge, the price was converted to thousands of leones (divided by 1,000). The price coefficient is negative and significant and the general directions of coefficients are unchanged. In terms of the constant term, the white coefficient is about twice as large as the constant, implying a more than doubling of the value. The pure black coefficient is even larger in this model than in the conditional logit, however. None of the other mixes of black and white are statistically significant. The standard deviations of the distributions of the pure white and pure black coefficients are also statistically significant, implying that there is significant variation in how respondents value these coat colors.

In the Appendix, we also estimate a latent class model (LCM) to understand whether certain groups are driving the results. Using three classes, we find that about 26% of the respondents value pure whiteness above all other coat colors. However, there is little evidence that any respondent characteristics can explain who is in this group relative to being in a group that does not value any coat colors other than the base one. We also present our estimates of the mixed logit in willingness-to-pay space. Estimated this way, the directions of the coefficients are the same but the magnitudes appear to be larger than in preference space.

Conclusion

The objective of our work was to determine how sheep-owning households in Sierra Leone value the coat color of their sheep. It has often been assumed that preferences for coat color are based on a productive outcome. For example, the documented penalty for black-coated sheeps has been assumed to be because these animals are less heat tolerant and may attract more flies. Yet, there has also been speculation that cultural preferences may drive some of the preferences for coat color.

Using scientific diagrams that vary coat color strategically, we find evidence that respondents valued pure white sheep at least double the value of the base category. Respondents did not value a sheep with only a few black spots but otherwise white as much as a pure white sheep, suggesting that there is a preference for sheep to be pure white. With respect to the reasons behind the valuation, we do not find any evidence that the religion of the respondent affects this value.

Our study has some shortcomings that might ideally be addressed in future work. This study used a convenience sample of households around Njala University but ideally a future study would be able to do a more random selection of households to ensure no sample selection in the results. Another shortcoming of this study, and of most choice experiments,

is hypothetical bias. We only had three respondents choose to opt out of choosing a sheep, which may be either because our prices were too low or respondents had hypothetical bias.

Our work makes a crucial contribution to the literature on the valuation of animals in low-income contexts. While socio-cultural factors are often mentioned as potential reasons for valuation of traits like coat color, there has been little work critically examining whether these can be explained by preferences for productive traits. We find evidence that coat color might, indeed, be valued for socio-cultural reasons in Sierra Leone. Future work should explore the robustness of these results in similar contexts where factors like religion and culture may be playing a role in how households value their livestock.

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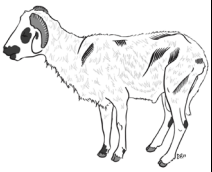
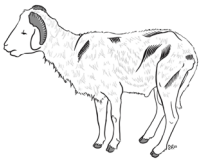
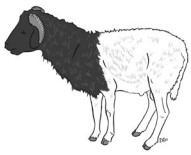
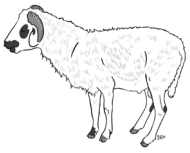
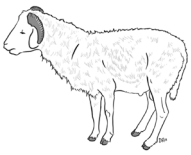
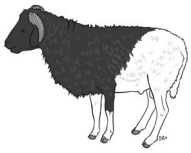
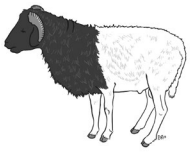
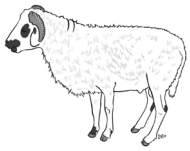
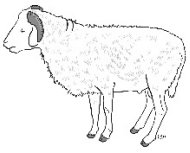
Appendix

Supplementary Figures

Figure A1: Weight and Color Combinations

	Color				
	Black	Black and White	White with Black Head	White with Black Spots	White
Below Average					
Average					
Above Average					

Latent Class Model

Table A1: Latent Class Model

	(1) Class1	(2) Class2	(3) Class3
Price	-0.0019***	-0.00005	0.0003*
	(0.0003)	(0.0002)	(0.0001)
Color			
White	2.8054***	1.5262***	-0.9230***
	(0.5801)	(0.3486)	(0.3047)
White with Black Spots	1.0322**	2.3857***	-2.3970***
	(0.4384)	(0.5326)	(0.4526)
Black and White	-0.8256*	0.8597**	-0.7911**
	(0.4536)	(0.3948)	(0.3334)
Black	-0.1394	-0.2787	-3.1192***
	(0.4519)	(0.4089)	(0.4194)
Weight			
Below Average Weight	-2.0371***	-1.8879***	-0.4708
	(0.5885)	(0.3820)	(0.2913)
Above Average Weight	-0.8688	0.5492	-0.1634
	(0.5926)	(0.3874)	(0.3258)
Class Share	26.6%	33.2%	40.2%
Observations	2,470		

Table A2: Membership Model

Base: Class 3	Class 1	Class 2
Education (Base: No Schooling)		
Some Primary Schooling	-2.3010*	0.5180
	(1.2264)	(0.7800)
Completed Primary or Some Secondary	-0.3210	-1.1606
	(0.8836)	(0.8551)
Completed Secondary or Some Post-Secondary	0.3762	-1.1156
	(1.0914)	(0.9561)
Age (Base: 30-40 years)		
< 30 years	-2.4617*	0.9255
	(1.3507)	(1.0895)
30-40 years	-1.9463	0.5422
	(1.2679)	(0.9170)
50-60 years	-1.5838	0.9222
	(1.1800)	(0.8720)
> 60 years	0.2298	-1.0920
	(1.0420)	(1.3536)
Female	-0.9314	0.5559
	(0.7690)	(0.5888)
Income (Base: 500k-1500k)		
< 500k Monthly	-3.3391**	0.3317
	(1.4641)	(0.8009)
1,000k – 1,500k Monthly	-2.8920**	1.3870*
	(1.2270)	(0.8077)
> 1,500k Monthly	-2.1059**	0.8596
	(1.0002)	(0.8135)
Base: Mende		
Temne	-0.0712	0.1874
	(0.8800)	(0.8081)
Fula	0.9790	2.8800**
	(1.6662)	(1.4114)
Other	1.8397	-16.9685
	(1.1217)	(1,467.7334)
Muslim (Base: Christian/Other)	1.7360	-0.8768
	(1.0807)	(0.7596)
Asset Index	1.3653***	1.1975**
	(0.4936)	(0.4886)
Reported Income from Livestock	-2.3070***	0.0013
	(0.8706)	(0.8021)
Number of Sheep Kept	0.0705	-0.1266*
	(0.0682)	(0.0764)
Keep Animals for Eid al-Adha	0.0217	1.2494*
	(0.6507)	(0.7009)

Mixed Logit, WTP Space

Table A3: Mixed Logit Results, WTP Space

		Estimate
Constant	Mean	0.513***
		(0.150)
	Std. Dev.	0.856***
		(0.211)
Price	Mean	-1.270***
		(0.389)
	Std. Dev.	4.142***
		(0.792)
Weight (Base: Avg. Weight)		
Below Avg. Weight	Mean	5.426***
		(0.038)
	Std. Dev.	5.395***
		(0.043)
Above Avg. Weight	Mean	-3.351***
		(0.048)
	Std. Dev.	-2.170***
		(0.021)
Color (Base: White, Black Head)		
White	Mean	-5.121***
		(0.037)
	Std. Dev.	-2.243***
		(0.017)
White with Black Spots	Mean	0.723***
		(0.026)
	Std. Dev.	2.771***
		(0.020)
Black and White	Mean	0.131***
		(0.017)
	Std. Dev.	3.011***
		(0.024)
Black	Mean	25.123***
		(0.231)
	Std. Dev.	21.785***
		(0.194)
Observations	2,730	

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$